

SUPPLY CHAIN EXCEPTION CONTROL SYSTEM

Risk-Priority Engine, Synthetic Operational Data, Independent Outcomes and Evaluation Specification

Version 1.0 | Frozen portfolio design baseline | 20 July 2026

CONTROL POSITION

The score is a deterministic risk-priority score, not a probability. Candidate Risk is analytical. A lower score never resolves an exception. Resolution and closure remain human-controlled in the MVP.

Prepared for	Portfolio project - simulated Australian mid-sized distributor/manufacturer
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Governing sources	Project Charter v1.0; Business Requirements Specification v1.0; Frozen Operational Exception Lifecycle and Operating Model v1.0
Primary grain	One purchase-order-line risk episode
Scope	Logical rules and generation/evaluation contracts only; no executable SQL, Python, UI, BI measures, notifications or AI implementation
Status	Frozen for physical data-model implementation, subject to the explicitly labelled portfolio assumptions

EVIDENCE LIMITATION

All thresholds, volumes, prevalence ranges, workload ranges and target metrics are proposed portfolio assumptions or acceptance targets. They are not production policies, validated forecasts, achieved performance or commercial results.

Document control and authority

Authority	Binding use
1. Project Charter v1.0	Business problem, objectives, scope, controls, success targets and limitations.
2. BRS v1.0	Requirement identifiers, deterministic 0-100 score, contribution evidence, initial bands, independent outcomes and acceptance gates.
3. Frozen Operating Model v1.0	Superseding lifecycle semantics: eight states, one active episode per line/site, human-controlled resolution/closure, Reopened as event, and linked successor after closure.
4. This specification	Frozen rule-policy, synthetic-data and evaluation assumptions before physical implementation. It may not redefine lifecycle decisions.

Decision labels

Label	Meaning
[GOV]	Directly preserved governing decision.
[PA]	Proposed portfolio assumption frozen for the synthetic MVP; not empirically validated.
[TEST]	Expected distribution or acceptance/control range to be measured after generation.
[DEFERRED]	Intentionally outside the MVP; not silently omitted.

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1. Risk framework overview

The engine assesses every eligible open purchase-order line at each successful run. It converts a limited set of observable operational signals into an ordinal 0-100 risk-priority score, preserves the result of every evaluated rule, and produces a Candidate Risk disposition before any lifecycle decision. It is a work-prioritisation control, not an estimate of the probability of delay, stockout or loss.

FROZEN ARCHITECTURE

Validated as-of facts -> rule evaluations -> capped dimension contributions -> mitigation -> 0-100 risk-priority score -> severity band -> candidate disposition -> open/link/block decision. Outcome truth is generated and retained through a separate contract and is never an input to scoring.

1.1 Binding control principles

- [GOV] Deterministic: identical inputs, as-of time, configuration and rule version produce identical outputs.
- [GOV] Explainable: every rule stores availability, observed value, comparator, threshold, trigger result, gross points, cap adjustment, applied points and explanation code.
- [GOV] Versioned: every score cites one immutable approved version and effective interval.
- [GOV] Missing data is an explicit state. A rule may be AVAILABLE-NOT-TRIGGERED, TRIGGERED, UNAVAILABLE or INVALID; UNAVAILABLE is never presented as safe.
- [GOV] Candidate Risk is analytical and never an operational lifecycle status.
- [GOV] A falling score, full receipt flag or supplier promise can create possible-resolution evidence but cannot move an episode to Resolved or Closed.
- [GOV] Partial receipts change residual exposure; they do not automatically end the episode.
- [GOV] Delay and stockout signals on the same stable PO line/site normally belong to one active episode.
- [GOV] Calculated score and operational override remain separate; an override cannot rewrite the original score.
- [PA] High-value and operational-critical orders receive an additional human-control opening guard without changing the calculated band.

Governing-source trace: Charter sections 8, 11, 13-16; BRS FR-005-010, DQ-007/009/010, RULE-001-014; Frozen Operating Model sections 1-4 and 12.2.

2. Risk dimensions

Dimension	Purpose	Rules	Cap / floor
Delivery and timing	Materialised or emerging lateness.	RPR-DLY-01 to 03	[PA] Maximum +40
Inventory exposure	Projected inability to cover demand before usable receipt.	RPR-INV-01 to 02	[PA] Maximum +30
Visibility and data uncertainty	Near-term missing confirmation or unusable core stock/demand evidence.	RPR-VIS-01; RPR-DAT-01	[PA] Maximum +15
Supplier performance	Sufficient-history reliability and recent deterioration.	RPR-SUP-01 to 02	[PA] Maximum +15
Consequence	Operational criticality and residual financial exposure.	RPR-CON-01 to 02	[PA] Maximum +15

Dimension	Purpose	Rules	Cap / floor
Verified mitigation	Approved coverage that objectively protects the full residual need.	RPR-MIT-01	[PA] Minimum -15

Dimension caps control correlated signals, but they do not prove independence. Gross triggers and cap reductions remain visible. The pre-cap theoretical maximum is 115; the final score is capped at 100. The only negative rule is verified mitigation, applied after positive dimension caps and before the final 0-100 floor/cap.

3. Detailed rule catalogue

This section freezes the catalogue structure and scoring policy. The complete rule-by-rule cards, including all requested fields, are consolidated in Appendix A. Section 4 provides the contribution summary used for operational review.

Rule family	Rules	Key policy
Delivery	DLY-01 overdue; DLY-02 ETA after need; DLY-03 lead-time variance	Use accepted as-of dates only; delivery family capped at 40.
Inventory	INV-01 negative projected balance; INV-02 material shortfall	Focal residual quantity and deterministic daily projection; capped at 30.
Visibility/data	VIS-01 missing confirmation; DAT-01 unavailable core coverage inputs	Unavailability is explicit and can contribute uncertainty points; capped at 15.
Supplier	SUP-01 low trailing OTIF; SUP-02 recent deterioration	Minimum history and exact OTIF definition required; capped at 15.
Consequence	CON-01 criticality; CON-02 residual value	Consequence affects priority, not claimed likelihood; capped at 15.
Mitigation	MIT-01 verified full mitigation	Evidence, approval, full coverage and effective dates required; -15 maximum.

4. Score contribution table

Rule ID	Trigger summary [PA where numeric]	Gross points	Family cap
RPR-DLY-01	Overdue residual quantity	+30	Delivery +40
RPR-DLY-02	Expected receipt after need date: 1-3d / 4+d	+10 / +20	Delivery +40
RPR-DLY-03	Expected lead time >120% baseline	+10	Delivery +40
RPR-INV-01	Projected available balance <0 before focal receipt	+25	Inventory +30
RPR-INV-02	Projected shortfall >=25% of residual quantity	+5	Inventory +30
RPR-VIS-01	Need within 7d and no valid confirmation/ASN/ETA	+15	Visibility/data +15
RPR-DAT-01	Core inventory/demand evidence unavailable or stale	+15 urgent / +5 other	Visibility/data +15
RPR-SUP-01	Trailing 12m OTIF <85%, n>=20	+15	Supplier +15

Rule ID	Trigger summary [PA where numeric]	Gross points	Family cap
RPR-SUP-02	Recent OTIF deterioration ≥ 10 percentage points	+5	Supplier +15
RPR-CON-01	Effective approved criticality flag	+5	Consequence +15
RPR-CON-02	Residual value A\$50k-<150k / $\geq$ A\$150k	+5 / +10	Consequence +15
RPR-MIT-01	Approved full mitigation covers residual requirement	-15	Mitigation floor -15

5. Missing-data treatment

NO SILENT ZERO

A numeric zero contribution is permitted only when a rule had sufficient valid inputs and its condition was false. If inputs are absent, stale, invalid or insufficient, the rule result is UNAVAILABLE or INVALID and the missing-signal reason is stored separately from points.

Class	Examples	Treatment	Scoring effect
A - structural mandatory	Stable keys; site/item/supplier references; ordered/received quantity; base UOM; need date; currency conversion where value is used.	Stop the affected publication or quarantine the affected record under the governing validation severity. No normal Candidate Risk is published as complete.	No fabricated score. Record INVALID and data-quality disposition.
B - core analytical	Inventory snapshot, demand forecast/firm demand, expected receipt, supplier history.	Continue only under a qualified score when permitted. Store UNAVAILABLE, missing code, source age and affected rule. Apply RPR-DAT-01 or RPR-VIS-01 only when its exact condition is met.	No automatic safe value; score confidence is QUALIFIED.
C - optional/mitigating	Substitute approval, protected stock evidence, nonessential supplier trend detail.	Do not infer mitigation or benefit. Store NOT-EVIDENCED or UNAVAILABLE.	No negative contribution; standard risk rules remain.

5.1 Score confidence status

Status	Exact use
COMPLETE	All Class A inputs valid and every rule-relevant Class B signal either available or legitimately not applicable.
QUALIFIED	Class A valid, but at least one rule-relevant Class B signal is unavailable/stale/insufficient. Score remains usable only with the displayed qualification.
NOT SCORABLE	Class A failure prevents a defensible score; record is stopped/quarantined according to the validation gate.

Scores are not rescaled to compensate for missing dimensions. Rescaling would imply that unavailable evidence has the same distribution as available evidence and would impair comparability. A separate missing-signal count, affected-dimension list and source-freshness summary accompany each score.

6. Severity-band rules

Calculated score [PA]	Calculated band	Candidate/workflow treatment
0-29	Monitor	Retain Candidate Risk. No score-based opening; a critical data-gap guard may still open a controlled-order episode.
30-49	Medium	Retain Candidate Risk. A controlled-order guard may open when need is near; otherwise monitor.
50-69	High	Qualifies for score-based opening if no active duplicate or valid blocking control exists.
70-100	Critical	Qualifies for score-based opening and critical human controls.

Boundary mapping is inclusive as shown: 29 remains Monitor; 30 starts Medium; 50 starts High; 70 starts Critical; 100 is the maximum. The band is derived from the calculated score and never overwritten. Any approved operational priority reduction is a separate, time-bound control record and does not change historical score evidence.

7. Opening and material-recurrence rules

7.1 New episode opening [PA]

Opening basis	Exact condition	Control
OPEN_SCORE	Calculated score ≥ 50 .	Automatic Open if eligible, no active episode with the logical key, and no valid candidate-blocking control.
OPEN_CONTROLLED_ORDER	Score 30-49; need date is within 14 calendar days inclusive; and either approved criticality is effective or residual line value $\geq$ A\$100,000.	Automatic Open with calculated Medium band retained. Human review is required; no synthetic severity inflation.
OPEN_CRITICAL_DATA_GAP	Need date is within 7 calendar days inclusive; controlled-order criterion above is met; and expected receipt, inventory or demand evidence is UNAVAILABLE.	Automatic Open regardless of score, with opening basis and missing codes visible.

The opening rules are workload assumptions, not validated operational policy. A lower score on later runs links to the active episode and may change current severity, but it never resolves, closes or deletes the episode.

7.2 Duplicate and recurrence controls

- [GOV] Active uniqueness key: source organisation/system + receiving site + stable PO ID + stable PO-line ID + active indicator. Risk type and split schedule are not key components in the MVP.
- [GOV] If Open, Assigned, Investigating, Action Agreed, Monitoring, Resolved or Suppressed exists for the key, the candidate links to it; a second active episode is prohibited.
- [GOV/PA] After Closed, a linked successor opens when current severity is High/Critical; or score is at least 15 points above the closing snapshot and crosses an applicable opening rule; or objective evidence reverses the closure basis.

- [GOV] Before Closed, suppression expiry/revocation, failed action, receipt reversal or invalidated evidence reactivates the same episode through a Reopened/Resolution Withdrawn event; it does not create a new state or duplicate episode.
- [GOV] Candidate evaluations continue while an episode is Suppressed. Valid suppression blocks duplicate opening but does not erase evidence or stop scoring.

8. Explanation-code catalogue

Code	Plain-language meaning	Source
DLY_OVERDUE	Residual quantity remains after accepted delivery date.	RPR-DLY-01
DLY_ETA_AFTER_NEED_1_3	Expected receipt is 1-3 days after operational need.	RPR-DLY-02
DLY_ETA_AFTER_NEED_4PLUS	Expected receipt is at least 4 days after operational need.	RPR-DLY-02
DLY_LT_VARIANCE_HIGH	Expected lead time exceeds baseline by more than 20%.	RPR-DLY-03
INV_STOCKOUT_BEFORE_RECEIPT	Projected available balance becomes negative before focal usable receipt.	RPR-INV-01
INV_SHORTFALL_MATERIAL	Minimum projected shortfall is at least 25% of residual quantity.	RPR-INV-02
VIS_CONFIRMATION_MISSING_NEAR_NEED	No valid accepted confirmation/ASN/ETA within 7 days of need.	RPR-VIS-01
DAT_CORE_COVERAGE_UNAVAILABLE	Inventory or demand evidence is missing, stale or unusable.	RPR-DAT-01
SUP_OTIF_BELOW_85	Trailing OTIF is below 85% with sufficient history.	RPR-SUP-01
SUP_OTIF_DETERIORATING	Recent OTIF has deteriorated by at least 10 percentage points.	RPR-SUP-02
CON_ITEM_CRITICAL	Effective approved service/production criticality applies.	RPR-CON-01
CON_VALUE_HIGH	Residual line value is A\$50,000 to below A\$150,000.	RPR-CON-02
CON_VALUE_VERY_HIGH	Residual line value is at least A\$150,000.	RPR-CON-02
MIT_FULL_COVERAGE_APPROVED	Approved substitute/protected stock fully covers residual need.	RPR-MIT-01
CAP_DELIVERY_40	Gross delivery contributions were capped at 40.	Calculation
CAP_INVENTORY_30	Gross inventory contributions were capped at 30.	Calculation
CAP_VISIBILITY_DATA_15	Gross visibility/data contributions were capped at 15.	Calculation
CAP_SUPPLIER_15	Gross supplier contributions were capped at 15.	Calculation
CAP_CONSEQUENCE_15	Gross consequence contributions were capped at 15.	Calculation
SIG_SUPPLIER_HISTORY_INSUFFICIENT	Supplier history is below the approved observation threshold.	Missing signal
SIG_LT_BASELINE_UNAVAILABLE	Item-supplier lead-time baseline has insufficient valid history.	Missing signal
SIG_SCORE_QUALIFIED	At least one rule-relevant core signal is unavailable.	Missing signal
OPEN_CONTROLLED_ORDER	Medium controlled order opened under human-control guard.	Opening
OPEN_CRITICAL_DATA_GAP	Controlled order opened because near-term core evidence is unavailable.	Opening
LINK_ACTIVE_EPISODE	Candidate linked to the existing active PO-line/site episode.	Disposition
BLOCK_VALID_SUPPRESSION	Candidate retained but duplicate opening blocked by effective suppression.	Disposition

Every triggered or unavailable rule carries one primary explanation code and may carry supporting cap/missing/disposition codes. Display wording is versioned; code meaning may not be silently repurposed.

9. Supplier-risk methodology

9.1 OTIF definition [PA]

Supplier OTIF is calculated at completed PO-line level over the trailing 12 months as of the score run. A line is On Time when the cumulative usable quantity reaches the In Full threshold no later than the latest accepted confirmed delivery date plus one calendar day. In Full means at least 98% of ordered quantity or residual shortage no greater than one base-UOM unit, whichever is more permissive. The line must satisfy both conditions to be OTIF-successful.

Treatment	Frozen rule
Observation threshold	At least 20 completed eligible lines for RPR-SUP-01. Below that, supplier reliability is UNAVAILABLE, not zero-risk.
Recent trend	Last 90 days requires at least 8 completions; preceding nine months requires at least 12. Deterioration is ≥ 10 percentage points and recent OTIF $< 90\%$.
Partial delivery	Success only if cumulative usable receipts meet In Full by the deadline. Later partials do not retroactively make the line on-time.
Cancellation	Supplier-caused cancellation is a failure; buyer/demand cancellation before supplier commitment is excluded; ambiguous cause is excluded and flagged.
Defect	Rejected quantity is not usable receipt until corrected/released. A defective on-time physical arrival may still fail In Full.
Corrections	Append-only receipt corrections restate the current metric prospectively for later runs; past stored score snapshots retain the metric version and inputs used.
Aggregation	Supplier overall is the MVP scoring level. Supplier-site/item-category views are retained for analysis but do not create additional score points.

Supplier risk intentionally carries a +15 family cap. Low OTIF and deterioration may both trigger, but the cap prevents the same performance history from adding +20. Both gross contributions remain explainable.

10. Inventory and stockout-exposure methodology

10.1 Deterministic projected available balance [PA]

For each item-site-day in the projection horizon, projected available balance (PAB) equals accepted on-hand at the as-of time, plus cumulative usable firm inbound receipts and approved transfers/substitutes due by that day, minus cumulative hard allocations, expected demand and the approved safety-stock reserve. The focal PO residual quantity is included only on its current expected usable-receipt date.

Element	Frozen treatment
As-of inventory	Latest successful item-site snapshot no older than 26 hours. Older or absent evidence makes the core coverage signal unavailable.
Demand	Firm allocations/customer or production requirements take precedence; remaining forecast is added without

Element	Frozen treatment
	double counting. Forecast source must be no older than 7 days.
Horizon	From score date through the later of need date or expected receipt plus 14 days, capped at 60 calendar days.
Stockout threshold	RPR-INV-01 triggers when any daily PAB is strictly below 0 before the focal usable receipt. Exactly 0 does not trigger.
Shortfall magnitude	RPR-INV-02 triggers when absolute minimum negative PAB divided by maximum(residual base-UOM quantity, 1) is $\geq 25\%$.
Safety stock	Subtracted once as a protected reserve; it is not separately scored. Site/item policy is effective-dated.
Substitutes	Included only with active approval, conversion ratio, available quantity and site/date applicability. The same coverage cannot be counted twice.
Missing UOM conversion	Class A invalid for the affected calculation; no PAB is fabricated.

PAB is a deterministic exposure model, not an inventory optimiser or probabilistic service-level model. Actual stockout outcomes are generated from realised demand, usable receipt events and hidden operational losses, not from the PAB rule.

11. Delay and delivery-risk methodology

11.1 Date hierarchy [GOV refined by PA]

Date	Purpose and precedence
Operational need date	When the item-site requirement must be protected. Stable historical value plus append-only amendments.
Latest accepted confirmed delivery date	Supplier promise explicitly accepted before the as-of cutoff. Used for overdue and OTIF. A later supplier proposal is not accepted merely because it exists.
Expected usable-receipt date	Latest valid accepted ETA/ASN adjusted for transport and quality hold; otherwise accepted confirmation; otherwise original expected receipt. Used for forward exposure.
Actual usable receipt	Outcome event after allocation and quality disposition. Never visible to an earlier score snapshot.

11.2 Delay rules

- Overdue is evaluated at Australia/Melbourne local date. The accepted delivery date itself is not overdue; the next local calendar day is.
- Expected-receipt-after-need tiers use calendar-day difference: 1-3 days adds 10; 4 or more adds 20; same-day/earlier adds 0.
- Expected lead-time variance compares order-to-expected-receipt calendar days with the median item-supplier actual usable lead time over trailing 12 months, minimum 10 completed observations. Exactly 120% does not trigger; greater than 120% does.
- Supplier proposals, confirmations and ASNs are observation signals subject to late reporting, optimism and correction. They are not outcome truth.

12. Order-value and product-criticality methodology

Control	Frozen treatment
Residual line value [PA]	Residual base-UOM quantity multiplied by order unit price, converted to AUD using the synthetic order-date monthly rate. A\$50,000 to <A\$150,000 adds 5; >=A\$150,000 adds 10.
Criticality [PA]	Item-site master has Standard, Service-critical or Production-critical. Either critical class adds 5 when effective and approved.
Governance	Criticality is effective-dated, site-scoped, manager-owned and never inferred from current stockout. Missing criticality is a master-data defect, not Standard.
Controlled order [PA]	Either critical class or residual value >=A\$100,000. Medium controlled orders within 14 days can open under OPEN_CONTROLLED_ORDER.
Human authority	Controlled orders require manager approval for material suppression, priority reduction, accepted residual risk and exceptional closure. Calculated score remains visible.
Interpretation limit	Value and criticality measure consequence/attention, not probability. A high-value order can be operationally safe and a low-value component can still cause a severe stockout.

13. Partial-receipt and split-schedule treatment

Situation	Frozen treatment
Residual quantity	Ordered base-UOM quantity minus net usable allocated receipts and approved cancellations, floored at zero. Receipt corrections/reversals change the net through append-only events.
Receipt allocation	Use a valid schedule reference when present; otherwise allocate earliest due open schedule first. Ties use stable schedule ID. Over-receipt above tolerance remains a defect/exception, not negative residual risk.
Split schedules	All schedules are score components within one PO-line/site episode. Show earliest threatened need date, schedule residuals and total exposure; schedule ID is not an MVP active-exception key.
Risk recalculation	Every run recalculates on residual quantity and current schedule facts. Fully protected schedules cease contributing, while remaining schedules may continue to trigger.
Resolution	A partial receipt can reduce severity or support mitigation. It does not resolve unless the selected human resolution category proves the protected requirement is fully covered.
Correction/reversal	Link to the original receipt, recompute residual exposure, create possible-resolution-reversal evidence, reactivate same episode before closure or open linked successor after material post-closure recurrence.

14. Rule-versioning method

Version element	Contract
Identifier	RPR-MAJOR.MINOR.PATCH, beginning RPR-1.0.0. Identifier is immutable once approved.
Major	Rule semantics, dimension structure, outcome interpretation or eligibility changes.
Minor	Points, thresholds, caps, bands, opening rules or approved input hierarchy changes.
Patch	Explanation wording/metadata correction proven not to change evaluated outputs.

Version element	Contract
Activation	Owner, rationale, test evidence, approval, effective-from time and prior-version reference required. Only one production-target version is active at an as-of time.
History	Historical scores retain original version and contribution set. Prospective recalculation never overwrites history.
Backtest	A different rule version may be applied to a frozen historical as-of dataset only as a separate labelled experiment; original operational snapshots remain unchanged.
Reproducibility	Version package includes rule catalogue, thresholds/caps, explanation dictionary, input-contract version and configuration checksum.

15. Candidate-disposition rules

Disposition	Exact use
MONITOR_BELOW_OPENING	Eligible/scored candidate does not meet any opening rule and no active episode exists.
OPENED_NEW_EPISODE	Opening rule met; uniqueness passed; no effective candidate block; new Open episode created.
LINKED_ACTIVE_EPISODE	Active episode exists for logical key; candidate and score snapshot link to it regardless of current score.
BLOCKED_VALID_SUPPRESSION	Matching episode is Suppressed or an approved candidate-level data/noise block is effective; candidate retained and no duplicate opens.
QUALIFIED_MISSING_SIGNAL	Score is usable but one or more core analytical signals are unavailable. This may coexist with another primary disposition.
NOT_SCORABLE_DATA_FAILURE	Structural mandatory data failed; no normal score/opening is asserted. Validation/quarantine evidence owns the record disposition.
CLOSED_NO_RECURRENCE	Closed predecessor exists but current candidate does not meet material recurrence; candidate retained without successor.
OPENED_LINKED_SUCCESSION	Closed predecessor exists and material recurrence condition is met; successor opens with predecessor link and new SLA clocks.

A candidate may carry one primary lifecycle disposition plus qualification codes. Dispositions are analytical decisions, not lifecycle states.

16. Synthetic organisation profile

SYNTHETIC ORGANISATION

MerriVale Supply & Manufacturing (Synthetic) is a fictional Australian mid-sized distributor/manufacturer. It has no relationship to a real company, supplier, customer or employee.

Attribute	Frozen synthetic profile [PA]
Sites	2 Victorian receiving sites using the Australia/Melbourne operating calendar: Harbourline Distribution Centre (58% of lines) and Ironbark Assembly & Service Centre (42%).
Suppliers	Exactly 120 active synthetic suppliers: 45 Victorian/local, 35 interstate, 40 international/import.
SKUs	Exactly 1,000 active SKUs across resale goods, components, MRO/consumables and packaging.
History	1 July 2024 to 30 June 2026 inclusive (24 months / 730 calendar days).
PO activity	Approximately 300 purchase orders per week; mean 2.2 lines per PO; expected 62,000-75,000 PO lines.
Open population	Normal-day median 1,400-1,600 open PO lines; 95% of normal days expected within 1,200-1,800.
Roles	Six buyer portfolios, two inventory planners, one supply-chain manager, site warehouse roles, one administrator and one audit reviewer; all identities synthetic.
Currency	AUD reporting currency; synthetic foreign-currency lines converted with frozen monthly synthetic rates.

17. Synthetic master-data specification

Entity	Grain	Expected volume [PA]	Required attributes	Control
Site	One site	2	Stable ID; name; region; calendar; operating function; active interval.	Exactly two; no missing calendar.
Supplier	One supplier	120	Stable ID; geography; route class; currency; category; onboarding date; active interval.	All synthetic; no real names.
Item	One SKU	1,000	Stable ID; category; base UOM; pack/MOQ; shelf/complexity class; price band; active interval.	Base UOM mandatory.
Item-site	One SKU at one site	1,600-1,800	Criticality; safety stock; reorder settings; planner; demand class; effective dates.	Criticality never null.
Supplier-item	One approved supplier-SKU link	3,000-4,500	Nominal lead time; price/currency; MOQ; allocation share; route; validity.	At least one source per active item-site.
Buyer mapping	One effective mapping rule	25-45	Supplier-site override; buyer-group/site; category/site; duty queue; effective dates.	Precedence is deterministic.
Calendar	One local date	730+ forward dates	Business day; opening/closing time; Victorian holiday; DST offset; version.	Mon-Fri 08:30-17:00 [GOV/PA].
FX rate	One currency-month	As required	Synthetic AUD conversion rate; month; source label; version.	No external market claim.

17.1 Master distributions [PA]

- SKU functional mix: 45% resale/finished goods, 35% components, 12% MRO/consumables and 8% packaging.
- Criticality: 83% Standard, 12% Service-critical and 5% Production-critical at item-site level; site-specific differences allowed.
- ABC value class: 20% A, 30% B, 50% C. XYZ demand class: 45% X, 35% Y, 20% Z.
- Supplier latent archetype for outcome generation only: 55% stable, 30% average, 10% volatile, 5% fragile. This field is hidden from the score engine.
- Nominal lead-time ranges: local 3-14 days, interstate 7-28 days, import 28-70 days; item/route variation is retained.

18. Purchase-order generation logic

1. Generate 104 weekly order volumes around 300 POs/week with seasonality, working-day placement and a 15% coefficient of variation; do not force exactly 300 every week.
2. Assign site, buyer portfolio and supplier from effective master relationships, with category and geography mix preserved.
3. Draw 1-6 lines per PO with mean approximately 2.2. Prevent duplicate stable line IDs and invalid item-supplier/site combinations.
4. Set quantity from expected 2-6 weeks of demand, rounded to pack/MOQ, with capped long-tail bulk orders. Keep ordered quantity in order UOM and a controlled base-UOM conversion.
5. Set operational need date from reorder timing, nominal supplier-item lead time and buffer. Retain original need date; later amendments are append-only events.
6. Generate supplier confirmation timing and accepted promise separately. A proposed revision becomes authoritative only when accepted in the observation stream.
7. Create 18-25% split-schedule lines with 2-4 schedules whose quantities reconcile exactly to the line quantity. Schedules remain components of one PO line.
8. Generate amendments for 6-10% of lines and cancellations for 1-3%, with buyer/supplier cause and time. No amendment rewrites opening facts.
9. Generate synthetic prices and currencies independently of risk outcome. Convert residual value with the order-date monthly synthetic FX rate.
10. Retain a run-independent stable PO/line identifier so daily candidate evaluations join to one logical obligation.

19. Inventory and demand generation logic

Process	Generation contract [PA]
Latent realised demand	Daily item-site demand follows class-specific processes: X stable/seasonal count demand; Y trend/seasonal demand; Z zero-inflated intermittent demand. Promotions, customer spikes and production changes are latent operational drivers.
Observed forecast	Created from lagged history plus bias and forecast noise. It is correlated with but not equal to realised demand. Some shocks are announced and visible; others remain hidden.

Process	Generation contract [PA]
Initial inventory	Set from 2-8 weeks of expected coverage by ABC/criticality class, with controlled random variation. Physical on-hand is never negative; unmet demand becomes backlog/lost-service truth.
Replenishment	Reorder-point/periodic review behaviour generates PO demand; planning noise creates early, adequate and late orders. This is not optimisation.
Daily movements	Issues, usable receipts, transfers, adjustments, quarantine and write-off events update item-site inventory. Every event has effective time and base UOM.
Snapshots	Daily end-of-business inventory snapshots for active item-sites, with rare controlled missing/stale extracts. Score as-of uses only the latest published snapshot.
Safety stock	Effective-dated item-site reserve derived from demand class and nominal lead time, then rounded. It is an operating assumption, not optimised policy.
Hidden shocks	3-5% of SKU-weeks receive promotion/production shocks; 35-45% are not visible in the forecast before onset, creating legitimate false negatives.

20. Supplier-performance generation logic

Supplier performance is generated from an underlying operational process, then observed OTIF is calculated from historical receipt truth. The future outcome is not drawn from the displayed OTIF threshold or points.

Layer	Method [PA]
Stable latent profile	Supplier baseline dispatch reliability, capacity buffer, promise optimism, quality propensity and route variability. Hidden from scoring.
Time-varying state	Monthly hidden state transitions among Normal, Strained and Disrupted. State persistence and common regional shocks create autocorrelation.
Order-specific variation	Item complexity, order size, route, congestion and independent random noise alter ship/receipt performance.
Observed commitments	Supplier promise and ETA are noisy, sometimes optimistic observations of planned performance, subject to reporting lag and correction.
Historical metric	Trailing OTIF and lead-time baselines are calculated only from events available before each score run; no future receipts leak into the feature.
New/small suppliers	Onboarding dates and low volumes create insufficient history. Their supplier rules are UNAVAILABLE rather than assigned average or zero risk.

21. Receipt, delay, defect and correction-event logic

Event family	Generation contract [PA]
Shipment/receipt	Actual dispatch and site arrival are drawn from latent supplier state, route and independent noise. Warehouse processing adds usable-receipt latency.
Partial receipt	15-25% of received lines have 2-4 receipt events. Valid schedule reference is present for most; missing reference invokes earliest-due allocation.
Delay	Lateness emerges when actual usable receipt exceeds the accepted promise/need benchmark; the generator never

Event family	Generation contract [PA]
	reads risk points or severity.
Defect/quality hold	2-4% of receipt events contain rejected/quarantined quantity. Usable quantity may be zero or partial until release/replacement.
Corrective shipment	Material defects may create a linked corrective inbound event with its own latency and cost/impact evidence.
Correction/reversal	0.5-1.5% of receipt events receive an append-only correction or reversal linked to the original. No cumulative receipt history is overwritten.
Cancellation	1-3% of lines cancel with buyer, supplier or data-correction cause. Cause controls OTIF inclusion and outcome interpretation.
Evidence timing	Source publication latency is generated separately from physical event time, allowing late facts and corrected facts without look-ahead.

22. Scenario-injection strategy

Scenario injection changes underlying operations or observation channels; it does not assign a risk score, severity or outcome label directly. Labels are derived after the events run.

Scenario	Scope	Underlying change	Evidence purpose
Port/route disruption	Selected import lanes; multi-week period	Longer transit, ETA revisions, correlated suppliers	Delivery and stockout TPs/FNs
Supplier capacity strain	Supplier/category cluster	Dispatch delays, partials, optimistic promises	Supplier signal and hidden-state overlap
Demand spike	Item-site/week cluster	Realised demand above forecast; some announced	Visible TP and hidden FN cases
ASN/confirmation outage	Observation channel	Missing/stale visibility while physical flow may remain normal	False positives and data guards
Warehouse congestion	Site/day cluster	Arrival-to-usable receipt delay	ETA error and stockout exposure
Quality defect batch	Supplier-item lot	Rejected receipt and corrective shipment	Receipt reversal/operational impact
Receipt correction	Recorded receipt subset	Reversal after apparent completion	Reactivation/successor cases
Forecast extract staleness	Data-run window	Core demand input unavailable	Qualified scores and human guard
Mitigation success/failure	Approved substitute/transfer cases	Variable actual effectiveness	Counterfactual vs realised outcomes
Site transfer obligation	Small PO-line subset	Receiving site changes with successor link	Episode-key negative test

22.1 Injection controls

- [PA] 80-85% of operational history is baseline stochastic behaviour; 15-20% is exposed to one or more named scenario windows. Scenario overlap is allowed but logged.
- Named boundary fixtures are generated separately from the main evaluation population and never included in accuracy metrics.
- At least 25% of scenario families/instances are held out from rule-threshold selection.

- Scenario registry stores scenario ID, latent/visible classification, affected entities/time, seed and intended test purpose; it does not store the desired score result.
- The generator must not rerun scenario seeds merely because recall or false-positive targets were missed.

23. Independent outcome-generation design

INDEPENDENCE CONTRACT

Operational truth is generated before scoring from separate seeds, hidden state and continuous/noisy processes. The outcome generator cannot read risk score, severity, rule ID, rule points, opening threshold, candidate disposition, lifecycle state or exception action when creating the baseline truth labels.

23.1 Generation sequence

1. Freeze master data, latent supplier profiles, latent demand processes and scenario registry.
2. Generate baseline operational truth: actual demand, dispatch, arrival, usable receipt, defects, corrections, stock position and unserved demand.
3. Derive an observation stream with reporting lag, missingness, accepted promises, forecast error and as-of snapshots. The score engine sees only this stream.
4. Generate Candidate Risk scores under the frozen rule version.
5. Optionally simulate human interventions and their stochastic effectiveness to produce realised outcomes. Baseline no-intervention outcomes remain immutable for detection evaluation.
6. Derive outcome labels from truth events using the frozen label definitions; publish labels only to the evaluation layer after the observation window closes.

23.2 Hidden drivers and controlled correlations [PA]

Outcome process	Hidden/non-score drivers	Permitted visible correlation
Actual lateness	Supplier hidden state, route shock, capacity load, item complexity, warehouse congestion, random transport error.	Trailing OTIF, promise/ETA and lead-time history correlate imperfectly with future lateness.
Stockout	Realised demand shock, usable receipt timing, shrinkage/quality hold, transfer availability, hidden substitution flexibility.	Forecast, PAB and item criticality correlate with exposure but do not determine truth.
Operational impact	Unserved quantity/duration, customer/production urgency, hidden flexibility, expedite options and random consequence noise.	Criticality and residual value influence consequence only; impact is not copied from their points.

[TEST] Desired correlations are quality checks, not forced exact values: visible trailing OTIF versus future late indicator point-biserial magnitude 0.15-0.35; expected receipt error versus actual receipt delay rank correlation 0.35-0.60; forecast versus realised weekly demand 0.50-0.80. Values outside these ranges require review, not automatic score tuning.

23.3 Frozen outcome labels [PA]

Label	Exact definition
MATERIAL_LATE	While residual quantity is open, cumulative usable receipt fails to reach the In Full threshold by the latest accepted confirmed date +1 calendar day; if no accepted confirmation exists, use need date. Late residual must exceed max(2% of ordered quantity, 1 base-UOM unit).
ATTRIBUTED_STOCKOUT	Actual unserved item-site demand is >0 while the line is open and before its usable receipt, and the line is the

Label	Exact definition
	earliest-due open inbound obligation whose residual quantity could have reduced the shortage. Same-date lines share attribution proportionally; any positive allocation yields the binary label.
ADVERSE_BASELINE	MATERIAL_LATE or ATTRIBUTED_STOCKOUT occurs in the immutable no-intervention truth path.
ADVERSE_REALISED	Same definition applied after simulated intervention effects. Used for operational learning, never as the sole detection-quality label.
IMPACT_LEVEL	None; Low (administrative/expedite only); Moderate (customer backlog or production reschedule); Severe (material service failure or production stop). Derived from realised unserved duration/quantity, hidden urgency/flexibility and noise.
OUTCOME_ONSET	First material late deadline breach or first attributed unserved-demand time, whichever occurs first.

23.4 Reproducibility and independence controls

- Use non-overlapping named seed domains for master data, operational truth, observation noise, data defects, scenario assignment and intervention effects. No seed is derived from rule version or score.
- Persist a truth-contract version and checksum before risk-rule calibration. Changing a score threshold or points must leave truth-event and outcome-label checksums unchanged.
- Outcome functions accept truth/master/scenario inputs only. A contract review rejects any score, exception or severity field in the baseline outcome interface.
- Keep locked months and held-out scenario instances inaccessible during threshold selection. Releasing them early invalidates the final test claim.
- Do not regenerate truth because metrics look weak. Regeneration is allowed only for documented structural failure such as broken volume, impossible chronology or unreconciled quantities, and requires a new dataset version.

24. Expected distributions and control totals

Measure	Expected value [PA]	Type	Control note
Sites	2	Exact	Master count
Suppliers	120	Exact	Master count; all synthetic
SKUs	1,000	Exact	Master count
History	730 calendar days	Exact	1 Jul 2024-30 Jun 2026
PO lines	62,000-75,000	[TEST] range	Must remain >=50,000
Normal-day open lines	Median 1,400-1,600	[TEST] range	95% normal days 1,200-1,800
Split-schedule lines	18-25%	[TEST] range	2-4 schedules; line quantities reconcile
Lines with partial receipts	15-25%	[TEST] range	Receipt allocations reconcile
Receipt defects	2-4% of receipt events	[TEST] range	Usable quantity differs from physical receipt
Receipt corrections/reversals	0.5-1.5%	[TEST] range	All linked append-only
Missing confirmation near need	5-9% of eligible near-need lines	[TEST] range	Do not force alert outcome
Qualified missing core signal	3-7% of candidate evaluations	[TEST] range	Includes scenario windows

Measure	Expected value [PA]	Type	Control note
MATERIAL_LATE	10-18% of completed lines	[TEST] diagnostic	Not tuned to pass risk KPI
ATTRIBUTED_STOCKOUT	4-8% of eligible lines	[TEST] diagnostic	Cluster-aware by item-site
ADVERSE_BASELINE	13-22% of eligible lines	[TEST] diagnostic	Union, not sum
Calculated Monitor	68-80%	[TEST] diagnostic	Candidate snapshot distribution
Calculated Medium	12-20%	[TEST] diagnostic	Includes controlled-order openings
Calculated High	6-10%	[TEST] diagnostic	Before operational override
Calculated Critical	2-5%	[TEST] diagnostic	Before operational override
New exception openings	25-50 per normal business day total	[TEST] workload assumption	Report per buyer and site; do not force

24.1 Mandatory reconciliations

- Source records = accepted + rejected/quarantined for every source/run.
- Eligible PO lines = scored Candidate Risk records for every successful run.
- Ordered schedule quantity = PO-line ordered quantity; allocated net usable receipts + residual + approved cancellation = ordered quantity within tolerance.
- One candidate per eligible PO line/run; no more than one active episode per logical line/site key.
- Candidate dispositions reconcile to monitor + opened + linked + blocked + closed/no recurrence + successor, with qualification as an overlay.
- Outcome labels reconcile to underlying truth events; score-rule changes do not change outcome checksums.

25. Evaluation metrics

25.1 Evaluation unit and alert definition [PA]

The primary unit is one eligible stable PO line/episode family, not each repeated daily score. An alert-positive line has at least one calculated High or Critical score before baseline outcome onset; for a no-adverse line, any High/Critical score before final completion/cancellation counts as alert-positive. A High/Critical score first occurring after adverse onset is a late detection, not a true positive, but remains workload.

Metric	Definition	Use / target
Recall / sensitivity	$TP / (TP + FN)$ using ADVERSE_BASELINE.	[PA target] $\geq 70\%$ for High/Critical alerts; disclose if missed.
Precision	$TP / (TP + FP)$.	How much alert workload corresponds to baseline adverse outcomes.
Alert false-positive proportion	$FP / (TP + FP)$. This is $1 - \text{precision}$.	[PA charter target] $\leq 35\%$. This is the charter formula; do not call it classical FPR without qualification.
Classical false-positive rate	$FP / (FP + TN)$.	Report separately for statistical clarity.
Specificity	$TN / (TN + FP)$.	Complements classical FPR.
Useful lead time	Melbourne business time from first High/Critical alert to outcome	[PA target] median ≥ 2 business days

Metric	Definition	Use / target
	onset and separately to need date.	where an observable signal existed.
Late-detection rate	Adverse lines first alerted at/after outcome onset / all adverse lines.	Shows alerts that were operationally too late.
Workload	New openings, active episodes and High/Critical candidates per day/buyer/site.	Compare with 25-50 total new openings/day assumption and timed-screening test.
Band outcome prevalence	Adverse-baseline share within Monitor/Medium/High/Critical.	Ordinal discrimination diagnostic only; not probability calibration.
PR-AUC / ROC-AUC	Ranking diagnostics using score against ADVERSE_BASELINE.	Optional secondary metrics; do not interpret score as calibrated probability.
Missingness gap	Recall/precision and opening rate for COMPLETE versus QUALIFIED scores.	Detects hidden safety or workload from missing data.
Segment stability	Metrics by site, supplier geography, criticality, ABC/XYZ, lead-time class and scenario family.	No segment claim when denominator is below 30; show counts.

25.2 Evaluation split and governance [PA]

Split	Months	Permitted use
Development	1-14	Validate generation, rules, boundaries and basic distributions. Outcome results may inform defect fixes, not final claims.
Threshold review	15-18	Assess workload and sensitivity alternatives; approve RPR-1.0.0. Freeze before locked test.
Locked validation	19-24	One primary performance evaluation under frozen dataset/outcome/rule versions. No retuning after results.
Scenario holdout	At least 25% of named scenario instances	Test robustness to unseen scenario instances/families; keep clusters wholly within one split.

Confidence intervals should be cluster-bootstrap estimates by supplier and item-site where feasible because outcomes are correlated. Repeated daily snapshots must not be treated as independent observations.

26. Boundary, negative and sensitivity scenarios

26.1 Mandatory boundaries

Boundary family	Fixtures
Date	Accepted date today / yesterday; ETA exactly 0, 1, 3, 4 days after need; need exactly 7 and 14 days away.
Score/band	0, 29, 30, 49, 50, 69, 70, 100; raw totals below 0 and above 100.
OTIF	n=19/20; 84.999%/85%; recent n=7/8; deterioration 9.999/10 percentage points.
Lead time	Baseline n=9/10; ratio 1.20 exactly and smallest representable value above 1.20.
Inventory	Minimum PAB 0 and smallest negative; shortfall ratio just below/at 25%; same-day receipt ordering.

Boundary family	Fixtures
Value	A\$49,999.99; A\$50,000; A\$99,999.99; A\$100,000; A\$149,999.99; A\$150,000.
Quantity	Zero residual; within In Full tolerance; one unit above tolerance; over-receipt; reversal to positive residual.
Missing/stale	Snapshot age exactly 26h and >26h; forecast exactly 7d and >7d; absent baseline/history.
Recurrence	Closing score +14/+15; successor score 49/50; objective closure reversal regardless score.

26.2 Negative controls

- A lower score, zero residual calculation, receipt flag or new promise must not automatically resolve or close an episode.
- Risk type, split schedule and repeated run must not create a second active exception for the same logical line/site.
- Insufficient supplier history must not apply 0% risk, average OTIF or the +15 rule.
- Missing mitigation evidence, expired approval or partial coverage must not subtract 15 points.
- An operational override must not alter calculated score, band, contributions or version.
- Future actual receipt, realised demand, later correction and outcome label must not appear in an earlier as-of score input.
- Changing score points, bands or opening threshold must not regenerate outcome truth or labels.
- Successful intervention must not convert a baseline true positive into a false positive; evaluate baseline and realised outcomes separately.

26.3 Sensitivity set [PA]

Parameter	Alternatives	Required comparison
Opening threshold	45, 50, 55	Recall, alert false-positive proportion, classical FPR, openings/day, buyer workload, lead time.
Critical band	65, 70, 75	Critical volume, controlled-order mix, missed severe impact.
OTIF threshold	80%, 85%, 90%	Supplier contribution prevalence, segment workload and detection metrics.
Stockout rule	PAB < 0; PAB < safety-stock floor	Inventory alert volume and adverse-outcome coverage; identify double counting.
Data uncertainty	+0 explicit only; +5/+15 v1; +10/+20	Missingness workload and missed controlled orders.
Dimension caps	Delivery 35/40/45; supplier 10/15	Contribution concentration, band migration and performance.
Value guard	A\$75k/A\$100k/A\$150k	Manual-control workload and severe-impact coverage.

Sensitivity results are evidence for later change control, not permission to choose the version that looks best on the locked test.

27. MVP simplification review

Retained in MVP	Simplified / deferred
Deterministic daily as-of scoring; complete contributions; explicit missingness; versioning; one active line/site episode; residual receipts; independent truth; human Resolved/Closed.	Intraday scoring, probabilistic forecasting, inventory optimisation, causal intervention estimation, route/production optimisation, separate schedule cases, learned models and real ERP integration.
Supplier OTIF plus recent trend with minimum history.	Complex supplier scorecards, Bayesian shrinkage, contract-specific OTIF and external risk feeds.
One deterministic PAB path using forecast, firm demand and approved inbound.	Monte Carlo stockout probability, multi-echelon allocation and constrained production planning.
Fixed AUD value tiers and three criticality classes.	Dynamic value-at-risk, margin/customer penalties and financial approval integration.
Named stochastic/injected scenarios with locked holdout.	Claiming the synthetic scenario frequencies represent a real company.

DO NOT SIMPLIFY AWAY

Missing-signal visibility, append-only receipt corrections, candidate-to-episode traceability, independent outcome seeds, score/outcome time separation, human resolution, duplicate prevention and honest target labelling are mandatory controls, not optional polish.

28. Frozen decision register

Decision ID	Area	Frozen decision	Status
RISK-001	Name and interpretation	0-100 ordinal risk-priority score; never probability.	GOV
RISK-002	Bands	Monitor 0-29; Medium 30-49; High 50-69; Critical 70-100.	PA frozen
RISK-003	Caps	Delivery 40; inventory 30; visibility/data 15; supplier 15; consequence 15; mitigation -15.	PA frozen
RISK-004	Missingness	Explicit AVAILABLE/TRIGGERED/UNAVAILABLE/INVALID; no rescaling; qualified score and uncertainty rule.	GOV/PA
RISK-005	Opening	Score ≥ 50 plus controlled-order and critical-data-gap guards.	PA frozen
RISK-006	Human control	Critical item or residual value $\geq A\$100k$ receives controlled-order guard and manager authority.	PA frozen
RISK-007	No score resolution	Falling score or objective signal creates evidence only; Resolved/Closed human-controlled.	GOV
EPI-001	Episode scope	One active stable PO-line/site episode covers delay and stockout; schedules are components.	GOV
EPI-002	Recurrence	Before closure reactivate same episode; after closure create linked successor under frozen material rule.	GOV/PA
SUP-001	OTIF	Trailing 12m; +1 day tolerance; 98%/1-unit In Full; minimum $n=20$; threshold $<85\%$.	PA frozen
SUP-002	Trend	90d vs preceding 9m; min 8/12; $\geq 10pp$ deterioration and recent $<90\%$.	PA frozen
INV-001	PAB	Deterministic daily item-site projection, negative-before-receipt rule, 60d maximum horizon.	PA frozen
DLY-001	Date hierarchy	Need; latest accepted confirmation; expected usable receipt; actual usable receipt.	GOV/PA

Decision ID	Area	Frozen decision	Status
VAL-001	Value tiers	A\$50k/+5; A\$150k/+10; human-control guard at A\$100k.	PA frozen
CRT-001	Criticality	Standard/service/production; effective item-site governance; critical +5.	PA frozen
RCV-001	Partial receipts	Append-only, schedule referenced or earliest-due allocation; score residual only.	GOV
MIT-001	Mitigation	Only full approved/evidenced coverage subtracts 15; partial coverage no mitigation points.	PA frozen
VER-001	Versioning	Immutable semantic version/effective interval; prospective activation; no history rewrite.	GOV
SYN-001	Scale	2 sites, 120 suppliers, 1,000 SKUs, >=50k lines/24m, normal ~1,500 open.	GOV baseline
SYN-002	Reference period	1 Jul 2024-30 Jun 2026; expected 62k-75k lines.	PA frozen
SYN-003	Scenario use	Scenario changes processes/observations, not target score/label; holdout >=25%.	PA frozen
OUT-001	Truth independence	Truth generated before scoring with separate seeds and no score/lifecycle inputs.	GOV
OUT-002	Counterfactual	Immutable baseline no-intervention outcome plus realised post-intervention outcome.	PA frozen
OUT-003	Labels	Material late, attributed stockout, adverse baseline/realised, impact and onset as defined.	PA frozen
EVAL-001	Unit	One stable PO line/episode family; repeated daily scores are not independent cases.	PA frozen
EVAL-002	Split	Months 1-14 development; 15-18 threshold review; 19-24 locked validation.	PA frozen
EVAL-003	False positive naming	Report charter alert false-positive proportion and classical FPR separately.	Frozen correction
EVAL-004	No accuracy claim	Performance remains unknown until data generated, locked and evaluated.	GOV

29. Critical self-review

Risk / weakness	Why it matters	Required treatment
Subjective weights	Points, thresholds and caps are expert-designed assumptions, not empirically estimated. A 70 score is not 70% risk.	Label all values [PA]; run sensitivity; change only through version control.
Correlated signals	Overdue, ETA lateness, lead-time variance, OTIF and PAB can describe the same underlying delay.	Dimension caps and contribution-correlation reporting reduce but do not eliminate double counting.
Missingness points	Adding uncertainty points mixes data-control urgency with operational exposure and may inflate workload.	Show score confidence/missing codes; compare explicit-only versus points sensitivity.
Simplified PAB	One deterministic path omits forecast distributions, allocation priorities, production constraints and multi-echelon effects.	Call it exposure projection; never claim optimal inventory or probability.
OTIF endogeneity	Visible historical OTIF and future delay share latent supplier behaviour. This is realistic correlation but can become overly tidy.	Use time-varying hidden states, order noise and held-out disruptions; audit correlation ranges.
Stockout attribution	Multiple inbound lines may contribute to one item-site shortage; earliest-due attribution is a simplification.	Report cluster-level item-site results and attribution sensitivity.
Intervention contamination	An effective action can prevent the realised adverse event, making a good alert appear false.	Use immutable baseline outcome for detection and realised outcome for intervention learning.

Risk / weakness	Why it matters	Required treatment
Synthetic role controls	Simulated approval roles cannot prove identity, security or real segregation of duties.	Describe control design only; do not claim enterprise security.
Scenario prevalence	Injected shocks can distort natural prevalence and make headline metrics unrepresentative.	Report baseline and injected-scenario metrics separately and show counts.
Designer leakage	One project author can knowingly or unknowingly tune outcome parameters to the score.	Freeze truth contract/checksum first; lock holdout; record every regeneration and reason.
High-value prioritisation	Financial value may crowd out low-value production-critical components.	Keep criticality separate; report missed severe impact by value tier.
No real validation	Synthetic acceptance cannot establish user adoption, policy validity, supplier behaviour or commercial benefit.	Limit claims to reproducibility, control behaviour and measured synthetic-test results.

FROZEN CONCLUSION

This specification is fit to guide logical/physical implementation and synthetic testing. It is not evidence that the rule engine is accurate, calibrated, production-ready or commercially beneficial. Those claims remain prohibited until generation, locked evaluation and, outside this portfolio, real operational validation occur.

Appendix A. Complete risk-rule catalogue

All numeric conditions and points below are [PA]. Every rule is evaluated for every eligible candidate unless marked not applicable; the result and availability are retained even when gross points are zero.

RPR-DLY-01 - Overdue residual quantity

Rule ID	RPR-DLY-01
Business rationale	Delay has already materialised while an obligation remains open.
Required inputs	As-of Melbourne date; stable PO line/site; residual base-UOM quantity; latest accepted confirmed delivery date; quantity tolerance.
Exact condition	Residual quantity > tolerance AND as-of local date is strictly later than the latest accepted confirmed delivery date.
Score contribution	+30 gross points.
Severity / cap	Delivery family cap +40; no automatic severity floor.
Missing-data behaviour	If confirmed date is absent, result UNAVAILABLE; RPR-VIS-01 may trigger near need. Missing need date/quantity is structural INVALID.
Explanation code	DLY_OVERDUE
Example	As-of 20 Jul; accepted date 19 Jul; 40 units residual -> +30.
Boundary test	As-of 19 Jul -> 0; as-of 20 Jul -> +30. Residual within tolerance -> 0.
Possible weakness	Calendar-day logic ignores intra-day delivery windows and accepted grace terms.

RPR-DLY-02 - Expected receipt after operational need

Rule ID	RPR-DLY-02
Business rationale	An accepted forward date beyond need reduces response time even before the order is overdue.
Required inputs	Operational need date; expected usable-receipt date; residual quantity.
Exact condition	Residual > tolerance AND expected receipt is 1-3 calendar days after need, or >=4 calendar days after need.
Score contribution	+10 for 1-3 days; +20 for 4+ days; mutually exclusive.
Severity / cap	Delivery family cap +40.
Missing-data behaviour	Expected receipt absent -> UNAVAILABLE; near-term missing visibility is assessed by RPR-VIS-01. Need date absent -> INVALID.
Explanation code	DLY_ETA_AFTER_NEED_1_3 or DLY_ETA_AFTER_NEED_4PLUS
Example	Need 22 Jul; expected usable receipt 26 Jul -> +20.
Boundary test	Same day -> 0; +1 and +3 -> +10; +4 -> +20.
Possible weakness	Date gap does not directly measure shortage magnitude or available inventory.

RPR-DLY-03 - Expected lead-time variance

Rule ID	RPR-DLY-03
Business rationale	An unusually long current cycle is an early warning against the same item-supplier history.
Required inputs	Order date; expected usable-receipt date; trailing 12m item-supplier median usable lead time; observation count.
Exact condition	At least 10 valid completed observations AND current expected lead time / baseline median >1.20.
Score contribution	+10 gross points.
Severity / cap	Delivery family cap +40.
Missing-data behaviour	Below 10 observations or absent expected date -> UNAVAILABLE with baseline code; no average or zero substituted.
Explanation code	DLY_LT_VARIANCE_HIGH; SIG_LT_BASELINE_UNAVAILABLE when unavailable.
Example	Expected 30 days; median baseline 20 days; ratio 1.50 -> +10.
Boundary test	Ratio exactly 1.20 -> 0; smallest value above 1.20 -> +10; n=9 unavailable, n=10 eligible.
Possible weakness	Median baseline may hide seasonality, route changes or structural supplier improvement.

RPR-INV-01 - Projected stockout before focal receipt

Rule ID	RPR-INV-01
Business rationale	A negative projected balance represents direct item-site service or production exposure.
Required inputs	Valid inventory snapshot; firm demand; forecast; safety stock; usable inbound/transfer/substitute schedule; focal expected receipt; residual quantity/UOM.
Exact condition	Minimum daily PAB is strictly <0 at any time before the focal usable receipt within the approved horizon.
Score contribution	+25 gross points.
Severity / cap	Inventory family cap +30.
Missing-data behaviour	Missing/stale inventory or demand -> UNAVAILABLE and RPR-DAT-01 assessment. Missing UOM -> INVALID.
Explanation code	INV_STOCKOUT_BEFORE_RECEIPT
Example	PAB falls from 12 to -8 units two days before receipt -> +25.
Boundary test	Minimum PAB 0 -> 0; minimum PAB -smallest unit -> +25.
Possible weakness	Deterministic forecast may overstate certainty and omits constrained allocation optimisation.

RPR-INV-02 - Material projected shortfall

Rule ID	RPR-INV-02
Business rationale	The scale of projected shortage increases operational attention beyond a small negative balance.
Required inputs	RPR-INV-01 result; minimum PAB; residual base-UOM quantity.

Rule ID	RPR-INV-02
Exact condition	RPR-INV-01 triggered AND $\text{abs}(\text{minimum negative PAB}) / \text{max}(\text{residual quantity}, 1) \geq 0.25$.
Score contribution	+5 gross points.
Severity / cap	Inventory family cap +30.
Missing-data behaviour	If INV-01 inputs unavailable/invalid, this rule inherits UNAVAILABLE/INVALID; it cannot trigger alone.
Explanation code	INV_SHORTFALL_MATERIAL
Example	Minimum PAB -30; residual 100; ratio 30% -> +5 in addition to INV-01.
Boundary test	24.999% -> 0; 25.000% -> +5.
Possible weakness	Residual-quantity denominator can make the ratio unstable for very small orders.

RPR-VIS-01 - Missing confirmation near need

Rule ID	RPR-VIS-01
Business rationale	Low supplier visibility near the requirement date reduces available response time.
Required inputs	As-of date; need date; residual quantity; accepted confirmation; valid ASN/ETA.
Exact condition	Residual > tolerance AND need date is from today through today+7 calendar days inclusive AND no valid accepted confirmation or ASN/ETA exists.
Score contribution	+15 gross points.
Severity / cap	Visibility/data family cap +15.
Missing-data behaviour	The missing confirmation is the trigger. Missing need date/quantity is INVALID rather than treated as this rule.
Explanation code	VIS_CONFIRMATION_MISSING_NEAR_NEED
Example	Need in 5 days; no accepted promise or ETA -> +15.
Boundary test	Need in 7 days -> +15; need in 8 days -> 0 but signal remains recorded as absent.
Possible weakness	A missing message may reflect integration failure rather than physical delivery risk.

RPR-DAT-01 - Core coverage evidence unavailable

Rule ID	RPR-DAT-01
Business rationale	Urgent work with unusable inventory/demand evidence must not appear safe.
Required inputs	Inventory snapshot presence/age; demand source presence/age; need date; residual quantity.
Exact condition	Residual > tolerance AND either inventory snapshot is absent/>26h old or demand evidence is absent/>7d old. Urgent tier when need is within 14 calendar days inclusive.
Score contribution	+15 urgent; +5 otherwise. One contribution even when both sources fail.
Severity / cap	Visibility/data family cap +15.
Missing-data behaviour	This rule explicitly represents permitted Class B missingness. If structural inputs fail, overall result is INVALID

Rule ID	RPR-DAT-01
	instead.
Explanation code	DAT_CORE_COVERAGE_UNAVAILABLE; SIG_SCORE_QUALIFIED
Example	Need in 10 days; inventory extract 31h old -> +15 and QUALIFIED score.
Boundary test	Age exactly 26h/7d is valid; any amount older is unavailable. Need in 14d urgent; 15d other.
Possible weakness	Uncertainty points mix data-quality urgency with operational exposure and may increase false alerts.

RPR-SUP-01 - Low trailing supplier OTIF

Rule ID	RPR-SUP-01
Business rationale	Persistent delivery failure is relevant when supported by adequate history.
Required inputs	Trailing 12m completed eligible supplier lines; frozen OTIF outcomes; count.
Exact condition	Observation count >=20 AND OTIF <85.00%.
Score contribution	+15 gross points.
Severity / cap	Supplier family cap +15.
Missing-data behaviour	Count <20 -> UNAVAILABLE with insufficiency code; no group average or zero risk.
Explanation code	SUP_OTIF_BELOW_85; SIG_SUPPLIER_HISTORY_INSUFFICIENT when unavailable.
Example	17 OTIF successes from 22 completions =77.3% -> +15.
Boundary test	84.999% -> +15; 85.000% -> 0; n=19 unavailable; n=20 evaluated.
Possible weakness	Supplier-wide OTIF may hide item, lane, contract or site differences.

RPR-SUP-02 - Recent supplier deterioration

Rule ID	RPR-SUP-02
Business rationale	A supplier can deteriorate before the full 12-month OTIF crosses 85%.
Required inputs	Last-90-day OTIF/count; preceding-nine-month OTIF/count.
Exact condition	Recent n>=8; prior n>=12; recent OTIF <90%; and prior OTIF - recent OTIF >=10 percentage points.
Score contribution	+5 gross points.
Severity / cap	Supplier family cap +15; cap adjustment visible if SUP-01 also triggers.
Missing-data behaviour	Either count below minimum -> UNAVAILABLE; does not affect SUP-01 if its own history is sufficient.
Explanation code	SUP_OTIF_DETERIORATING
Example	Prior 94%; recent 75%; decline 19pp -> +5 gross.
Boundary test	Decline 9.999pp -> 0; 10.000pp and recent <90% -> +5.
Possible weakness	Short windows are volatile; point estimate lacks uncertainty shrinkage.

RPR-CON-01 - Approved operational criticality

Rule ID	RPR-CON-01
Business rationale	A failure affecting service-critical or production-critical items has greater consequence.
Required inputs	Effective item-site criticality; approval/owner metadata; as-of date.
Exact condition	Criticality is Service-critical or Production-critical and effective at the as-of time.
Score contribution	+5 gross points.
Severity / cap	Consequence family cap +15; controlled-order human guard also applies.
Missing-data behaviour	Missing/invalid criticality is a master-data INVALID condition, not Standard.
Explanation code	CON_ITEM_CRITICAL
Example	Production-critical component at Ironbark site -> +5.
Boundary test	Effective-from instant inclusive; expired record not applied; overlapping records invalid.
Possible weakness	A static flag can become stale and may not represent customer-specific or time-specific consequence.

RPR-CON-02 - Residual order value

Rule ID	RPR-CON-02
Business rationale	Material financial exposure warrants additional attention and human control.
Required inputs	Residual base quantity; unit price; currency; order-date synthetic monthly AUD rate.
Exact condition	Residual AUD value $\geq 50,000$ and $< 150,000$, or $\geq 150,000$.
Score contribution	+5 for A\$50k-<A\$150k; +10 for $\geq$ A\$150k.
Severity / cap	Consequence family cap +15; human-control guard begins at A\$100k.
Missing-data behaviour	Missing price, currency or conversion makes value rule UNAVAILABLE/INVALID according to source criticality; no zero value assumed.
Explanation code	CON_VALUE_HIGH or CON_VALUE_VERY_HIGH
Example	Residual value A\$175,000 -> +10; controlled-order guard applies.
Boundary test	A\$49,999.99 -> 0; A\$50,000 -> +5; A\$149,999.99 -> +5; A\$150,000 -> +10.
Possible weakness	Gross order value does not measure margin, service cost, replacement difficulty or cash recoverability.

RPR-MIT-01 - Verified full mitigation

Rule ID	RPR-MIT-01
Business rationale	Objective, approved coverage can reduce residual consequence without erasing original exposure.
Required inputs	Approved substitute/protected-stock/transfer evidence; coverage quantity and conversion; effective

Rule ID	RPR-MIT-01
	site/date; residual requirement; safety stock.
Exact condition	Evidence is active and approved AND coverage protects 100% of the residual requirement and required safety stock through expected receipt.
Score contribution	-15 points after positive dimension caps.
Severity / cap	Mitigation floor -15; final score floored at 0.
Missing-data behaviour	Missing/expired evidence or <100% coverage -> not triggered; no negative points. Do not infer mitigation from a note.
Explanation code	MIT_FULL_COVERAGE_APPROVED
Example	Approved substitute covers all 80 residual units plus reserve through receipt -> -15.
Boundary test	99.999% coverage -> 0 mitigation points; 100% -> -15; expiry instant ends future application.
Possible weakness	Binary full-coverage rule ignores credible partial benefit and evidence quality differences.

Appendix B. Score calculation specification

For an eligible Candidate Risk evaluation under one rule version:

1. Evaluate all rules and record availability/result before applying caps.
2. For each positive family, sum gross triggered points and apply the family cap. Record any reduction as a separate CAP explanation contribution.
3. Positive subtotal = capped delivery + capped inventory + capped visibility/data + capped supplier + capped consequence.
4. Mitigation subtotal = -15 only when RPR-MIT-01 is triggered; otherwise 0.
5. Raw total = positive subtotal + mitigation subtotal.
6. Risk-priority score = minimum(100, maximum(0, raw total)). Integer points require no probabilistic rounding.
7. Derive calculated band from Appendix/Section 6 boundaries.
8. Attach score confidence, missing-signal list, source freshness, rule version, run ID and as-of timestamp.
9. Apply opening/deduplication/recurrence policy after scoring. Do not use lifecycle outcome as a score input.

WORKED EXAMPLE

Overdue +30; ETA 4+ days after need +20; lead-time variance +10 gives delivery gross 60, capped to 40 with CAP_DELIVERY_40=-20. Projected stockout +25 and material shortfall +5 gives inventory 30. Low OTIF +15 and deterioration +5 gives supplier gross 20, capped to 15. Criticality +5 and residual value A\$175k +10 gives consequence 15. No data/visibility points. Full approved mitigation -15. Final score = 40+30+15+15-15 = 85, Critical. The score is not an 85% probability.

Appendix C. Explanation-code catalogue

The complete code catalogue appears in Section 8. The implementation contract is: codes are stable identifiers; wording is versioned; one primary code is retained per triggered/unavailable rule; cap, qualification, opening and disposition codes may coexist; historical wording/version remains reproducible.

Code family	Required evidence fields
Rule trigger	Rule ID/version; availability; observed value/unit; operator; threshold; trigger; gross/applied points.
Cap	Dimension; gross subtotal; cap; reduction; contributing rule IDs.
Missing signal	Source/field; availability reason; observed source age; allowed freshness; affected rules; qualification status.
Opening/disposition	Opening rule/basis; active-key result; predecessor/current episode; suppression/block reference.

Appendix D. Synthetic data-generation contract

Domain	Binding contract
Organisation/master	Exactly 2 sites, 120 suppliers and 1,000 SKUs; all identifiers/names synthetic; effective-dated item-site, supplier-item, owner, criticality, calendar and FX data.
Operational period	24 months: 1 Jul 2024-30 Jun 2026; all event/effective times retained; Australia/Melbourne operational view.
PO population	>=50,000 lines; target 62,000-75,000; stable PO/line/site identity; 18-25% split schedules; amendments/cancellations append-only.
Inventory/demand	Daily item-site truth movements and snapshots; separate latent realised demand and observed forecast; no future leakage.
Supplier/receipts	Latent supplier states; noisy commitments; actual shipment/usable receipt; partials, defects, corrections and cancellations.
Data defects	Controlled missing/stale/invalid events with scenario ID and expected validation treatment; structural failures excluded from normal score claims.
Reproducibility	Dataset contract version; named non-overlapping seed domains; master/event/control totals and checksums; regeneration reason log.
Privacy	No real supplier, customer, employee, commercial price or confidential document. Synthetic evidence references only.

Appendix E. Independent outcome-generation contract

Contract clause	Binding requirement
OUT-C01 Order	Generate baseline truth before observations, scoring and workflow.
OUT-C02 Interface	Baseline generator cannot accept score, band, rule, candidate, exception, status or action fields.
OUT-C03 Seeds	Truth/observation/scenario/intervention seed domains are distinct and not derived from rule version.
OUT-C04 Noise	Include order-specific random variation, hidden supplier states, latent demand shocks, route/site disruptions and hidden flexibility.
OUT-C05 Correlation	Allow controlled correlation through shared real-world drivers, but never reuse score thresholds/points as label conditions.
OUT-C06 Labels	Derive material late, attributed stockout and impact from actual truth events under frozen definitions.
OUT-C07 Counterfactual	Retain baseline no-intervention and realised post-intervention outcomes; detection uses baseline.
OUT-C08 Immutability	Score changes leave truth/label checksums unchanged. Outcome regeneration requires documented structural reason and new version.
OUT-C09 Blinding	Keep months 19-24 and held-out scenarios locked until RPR-1.0.0 approval.

Appendix F. Expected distribution table

Section 24 is the complete expected distribution and control-total table. These ranges test whether the synthetic organisation is plausible and technically usable; they must not be tuned to guarantee the risk-performance targets. Out-of-range diagnostic outcomes require explanation and, only for documented structural implausibility, a new dataset version.

Appendix G. Evaluation plan

1. Validate data contracts, control totals, chronology, UOM, key uniqueness and reproducibility before any detection metric.
2. Hand-calculate all rules and boundaries; prove score decomposition, caps, floors, bands, missingness and version history.
3. Freeze RPR-1.0.0 after the threshold-review split; release locked validation only once.
4. Evaluate at PO-line/episode-family level against ADVERSE_BASELINE; report realised outcomes separately.
5. Publish confusion matrix, recall, precision, alert false-positive proportion, classical FPR, specificity, useful lead time, late detection and workload.
6. Stratify by site, supplier geography, criticality, value, ABC/XYZ, missingness, scenario family and outcome type, with counts and minimum-denominator warnings.
7. Run boundary, negative, leakage and outcome-invariance tests.
8. Run sensitivity only as pre-registered alternatives; do not select on locked-test performance.
9. Document failed targets honestly and retain the evidence package, versions, seeds and limitations.

Appendix H. Frozen decision register

Section 28 is the binding frozen decision register. Changes require owner, rationale, impact assessment, new version/effective date and retest of affected rule, data, outcome, lifecycle and reporting controls. A rule-policy change never authorises a lifecycle semantic change.

Appendix I. Critical risks of circular synthetic validation

Risk ID	Circularity mechanism	Mandatory control
CIR-01 Direct copy	Outcome label is created from score, points, band or opening result.	Prohibit score/lifecycle fields from baseline outcome interface; contract review.
CIR-02 Threshold reuse	Outcome generator uses <85% OTIF, PAB<0 score projection, 7-day visibility or score bands as truth probability steps.	Use continuous latent/noisy processes; labels only from actual truth events.
CIR-03 Generation order	Outcomes or adverse scenarios are assigned after seeing alerts.	Generate and checksum truth before observation/scoring.
CIR-04 Seed shopping	Regenerate until recall/false-positive targets pass.	Immutable dataset version; regeneration only for documented structural failure.
CIR-05 Parameter tuning	Tune hidden effects/correlations to reward chosen rules.	Pre-register correlation ranges and freeze outcome parameters before rule calibration.
CIR-06 Leakage	Earlier score snapshot receives actual receipt, future forecast, later correction or final OTIF information.	Strict as-of joins and leakage tests at every source field.
CIR-07 Intervention bias	Successful mitigation makes a valid alert look false.	Primary detection label is immutable baseline; realised outcome is secondary.
CIR-08 Duplicate snapshots	Daily repeated scores are counted as independent cases, narrowing intervals and inflating sample size.	Evaluate one PO line/episode family; cluster inference by supplier/item-site.
CIR-09 Scenario leakage	Same shock cluster appears in calibration and test through different lines.	Split entire scenario/supplier/item-site clusters, not individual rows.
CIR-10 Metric relabelling	FP/(all alerts) is called classical FPR, or score is described as calibrated likelihood.	Report denominators explicitly; use risk-priority language only.
CIR-11 Prevalence engineering	Injected scenario prevalence makes KPI easy and is presented as natural business incidence.	Separate baseline/scenario results and disclose prevalence/counts.
CIR-12 Hidden manual repair	Records that hurt results are excluded without predeclared data-quality reasons.	Every source record receives a reconciled disposition; exclusions pre-specified and counted.

FINAL EVIDENCE RULE

No predictive-accuracy, recall, precision, false-positive, lead-time or workload result may be claimed until the reference data has been generated, the locked outcome and rule versions have been evaluated, denominators have been reconciled, and failures/limitations have been retained.

Source references

- Supply Chain Exception Control System - Project Charter, Version 1.0, 19 July 2026.
- Supply Chain Exception Control System - Business Requirements Specification, Version 1.0, 19 July 2026.
- Supply Chain Exception Control System - Frozen Operational Exception Lifecycle and Operating Model, Version 1.0, 20 July 2026.